A project report on

CAPSTONE ANALYSIS :Customer Churn Prediction & Insights

THE DEVELOPER ARENA

Submitted by

Alka kushwaha

Student

Aks University Satna

Submitted to

Namanand Maske

Founder & CEO

www.thedevelopersarena.com

Project Overview

This project focuses on analyzing customer churn data to understand why customers leave a service. The main objective is to identify key factors influencing churn and build a predictive model that helps businesses reduce customer loss. By using data analysis and machine learning techniques, meaningful insights are generated to support decision-making.

Objectives

Analyze customer data to identify churn patterns

Perform data cleaning and preprocessing

Visualize important trends using graphs

Build a machine learning model for prediction

Provide business recommendations

Setup Instructions

Step 1: Install Required Libraries

Run the following command:

```
pip install pandas numpy matplotlib seaborn scikit-learn
```

Step 2: Open Google Colab

Go to Google Colab

Upload the dataset (customer_churn.csv)

Step 3: Run the Notebook

Execute all cells step-by-step

View outputs and graphs

Code Structure

```
Capstone_analysis.ipynb   # Main analysis code
Data/customer_churn.csv  # Dataset
Reports/report.pdf      # Project report
README.md               # Project overview
Requirements.txt        # Libraries used
```

```python
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns

# ⚠️ Load Data
df = pd.read_csv('/content/customer_churn (2).csv')

# ⚠️ Basic Info
print("Shape:", df.shape)
print(df.info())
print(df.describe())

# ⚠️ Data Cleaning
df.dropna(inplace=True)

# The 'Churn' column is already numerical (0 or 1) in the loaded data.
# Assign it directly to 'Churn_numeric' for model training.
df['Churn_numeric'] = df['Churn']

# Convert 'Churn_numeric' to string for plotting purposes.
df['Churn'] = df['Churn_numeric'].astype(str) # Use this 'Churn' for plotting

# 📊 ------------------ EDA ------------------

# 1. Churn Distribution
plt.figure()
sns.countplot(x='Churn', data=df)
plt.title("Churn Distribution")
plt.show()
plt.close()

# 2. Gender vs Churn - Fixed: Replaced 'gender' with 'PaperlessBilling' as 'gender' column d
plt.figure()
sns.countplot(x='PaperlessBilling', hue='Churn', data=df)
plt.title("PaperlessBilling vs Churn")
plt.show()
plt.close()

# 3. Contract vs Churn
plt.figure()
sns.countplot(x='Contract', hue='Churn', data=df)
plt.xticks(rotation=30)
plt.title("Contract vs Churn")
plt.show()
plt.close()

# 4. Monthly Charges
plt.figure()
sns.boxplot(x='Churn', y='MonthlyCharges', data=df)
plt.title("Monthly Charges vs Churn")
plt.show()
plt.close()

# 5. Tenure vs Churn
plt.figure()
sns.boxplot(x='Churn', y='Tenure', data=df)
plt.title("Tenure vs Churn")
plt.show()
plt.close()

# 📊 Correlation
plt.figure(figsize=(8,5))
sns.heatmap(df.corr(numeric_only=True), annot=True)
plt.title("Correlation Heatmap")
plt.show()
plt.close()

# ⚠️ ------------------ ML MODEL ------------------

from sklearn.model_selection import train_test_split
from sklearn.linear_model import LogisticRegression
from sklearn.metrics import accuracy_score, confusion_matrix

# Separate target variable (y) and features (X) BEFORE one-hot encoding
y = df['Churn_numeric']

# Drop CustomerID, the original 'Churn' (string for plotting), and the numerical 'Churn_nume
X = df.drop(columns=['CustomerID', 'Churn', 'Churn_numeric'])

# Convert categorical features in X using one-hot encoding
X = pd.get_dummies(X, drop_first=True)
```

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X = pd.get_dummies(X, drop_first=True)

X_train, X_test, y_train, y_test = train_test_split(X, y, test_size=0.2)

model = LogisticRegression(max_iter=1000)
model.fit(X_train, y_train)

y_pred = model.predict(X_test)

print("Accuracy:", accuracy_score(y_test, y_pred))
print("Confusion Matrix:\n", confusion_matrix(y_test, y_pred))
```

Visual Documentation

The project includes the following visualizations:

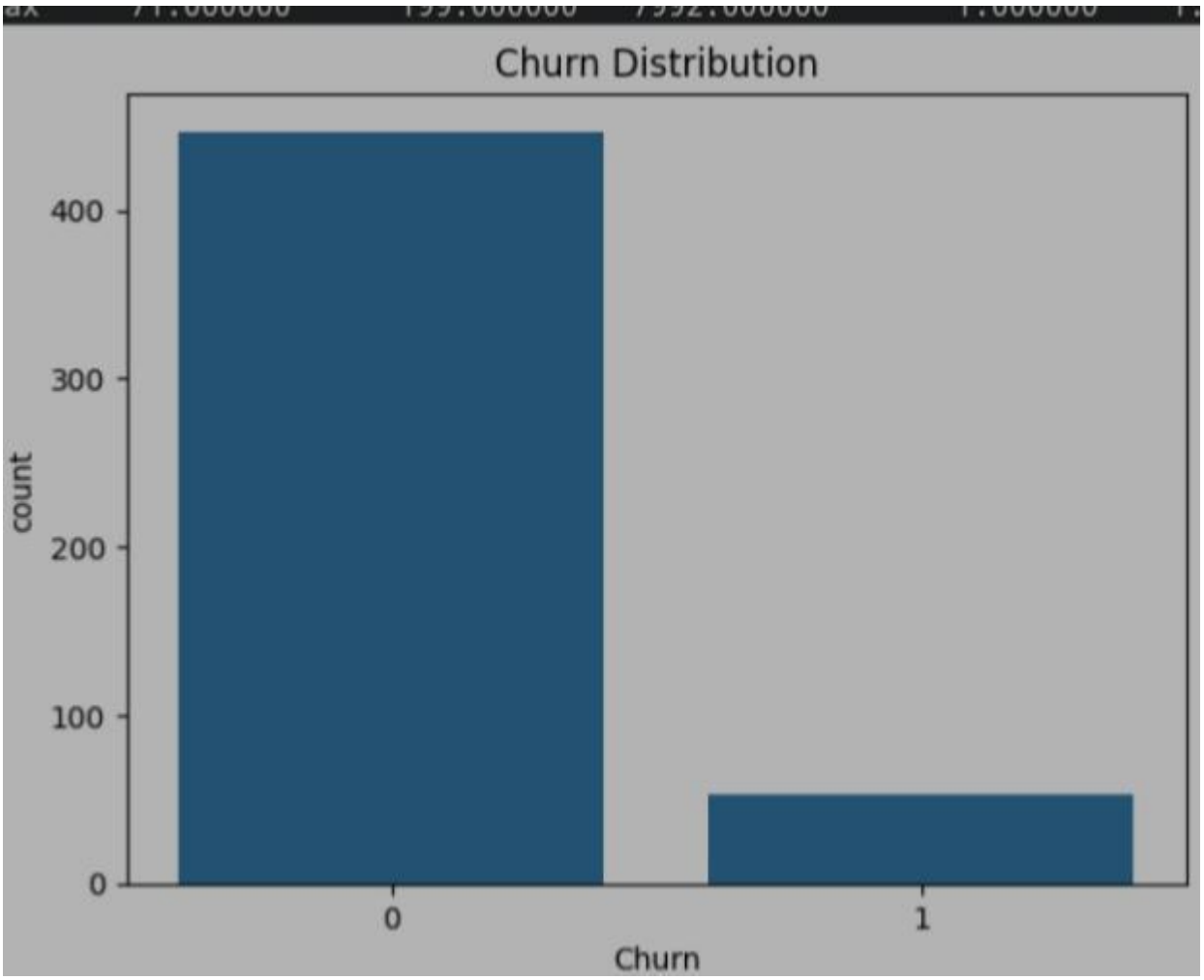
Churn distribution (Count Plot)

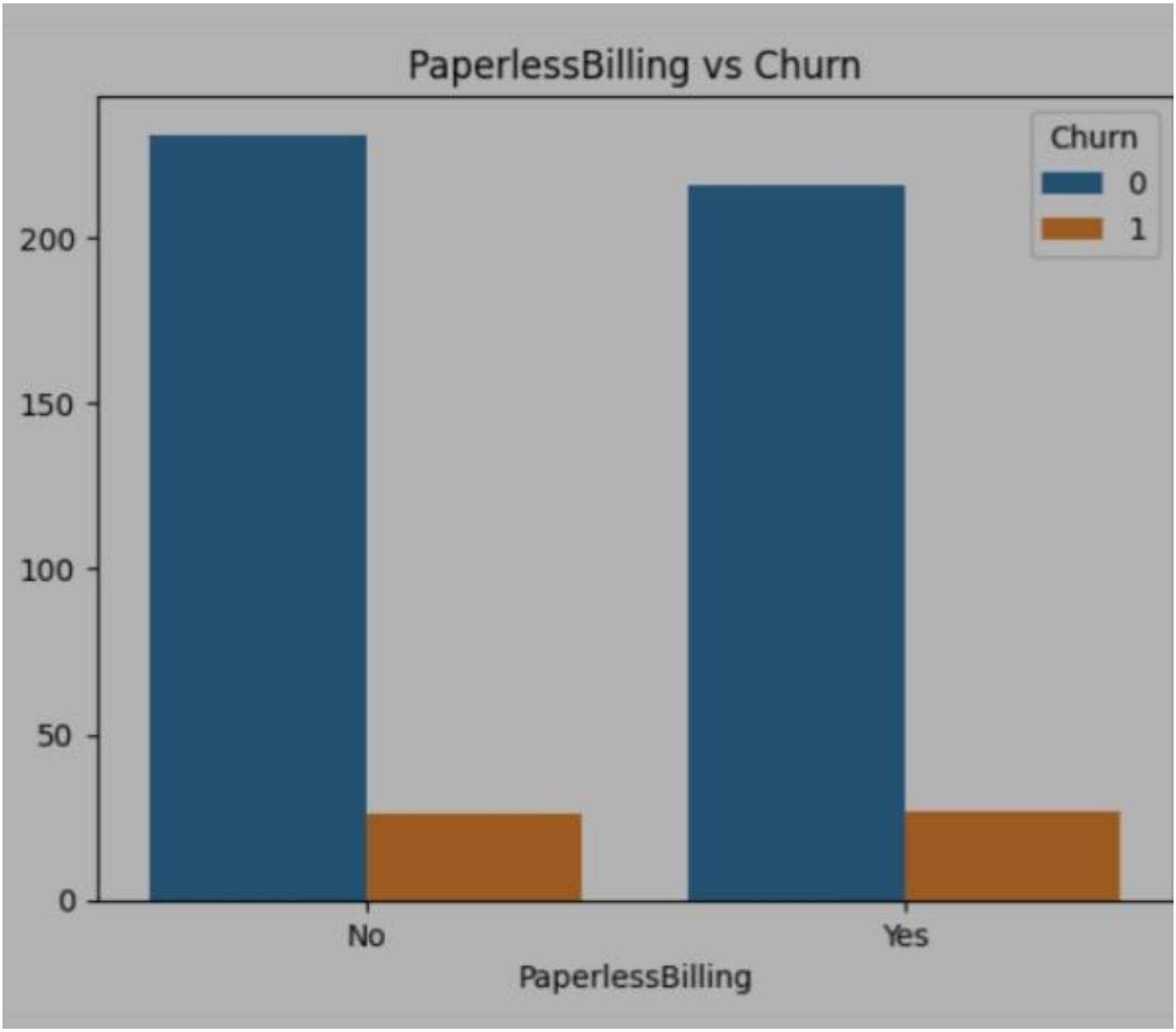
Gender vs Churn comparison

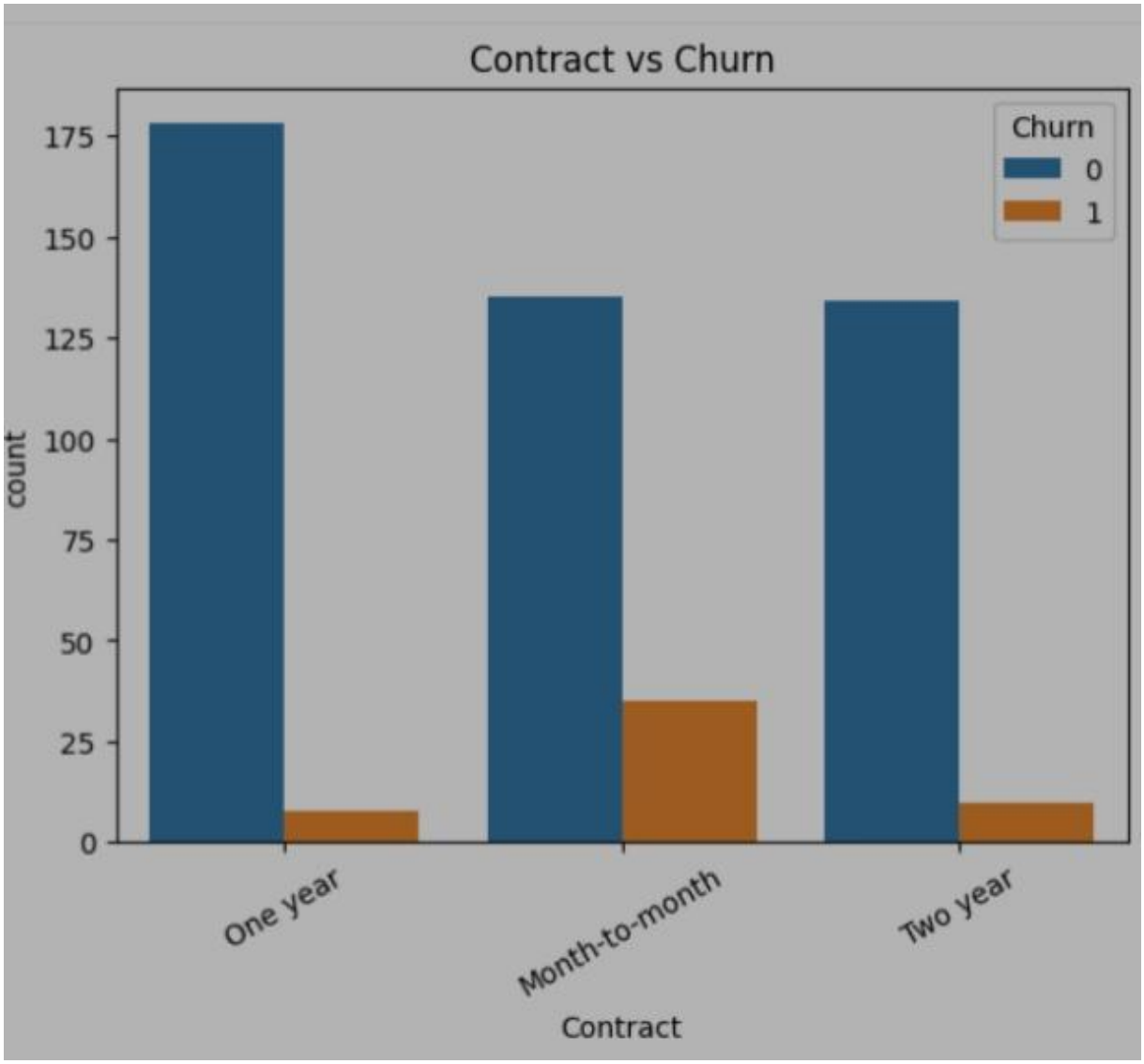
Contract type vs Churn

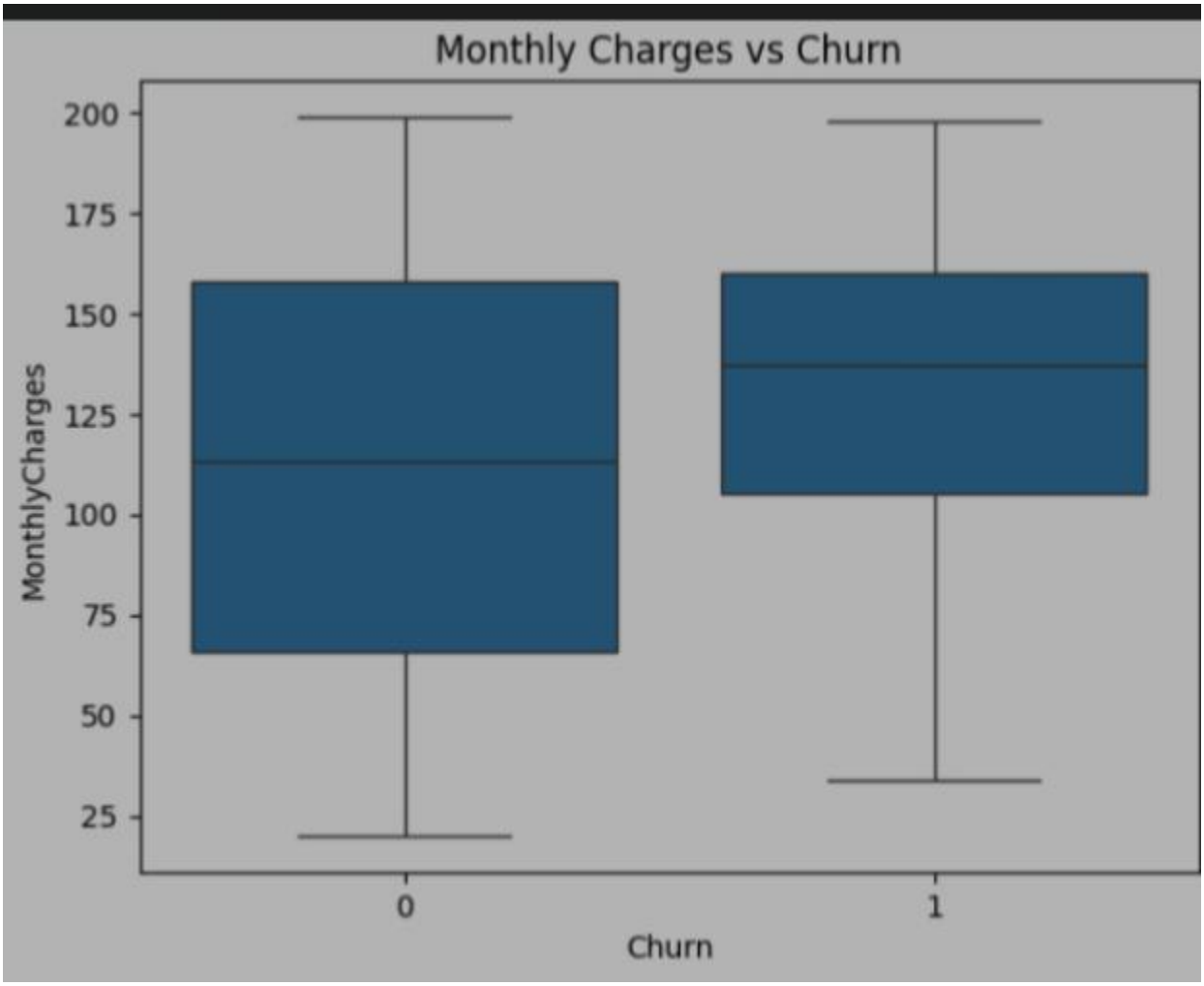
Monthly charges vs Churn (Box Plot)

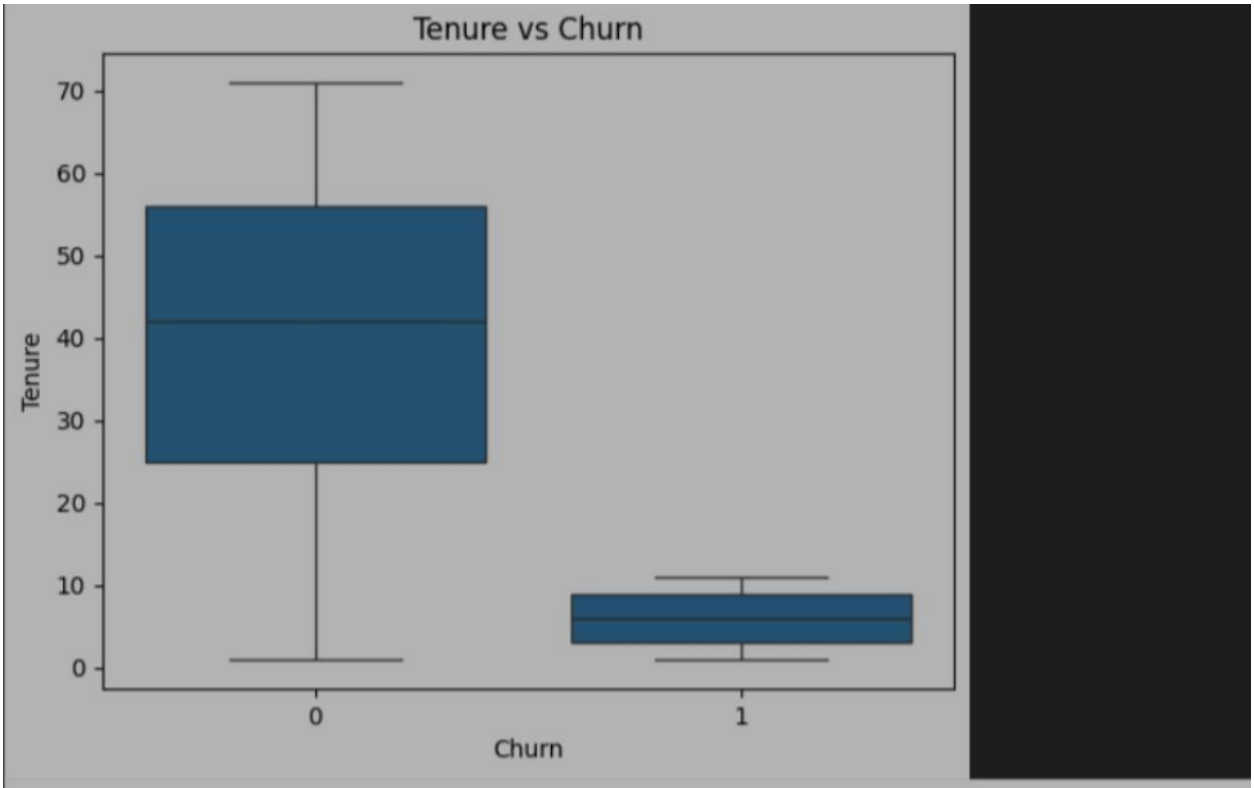
Correlation Heatmap

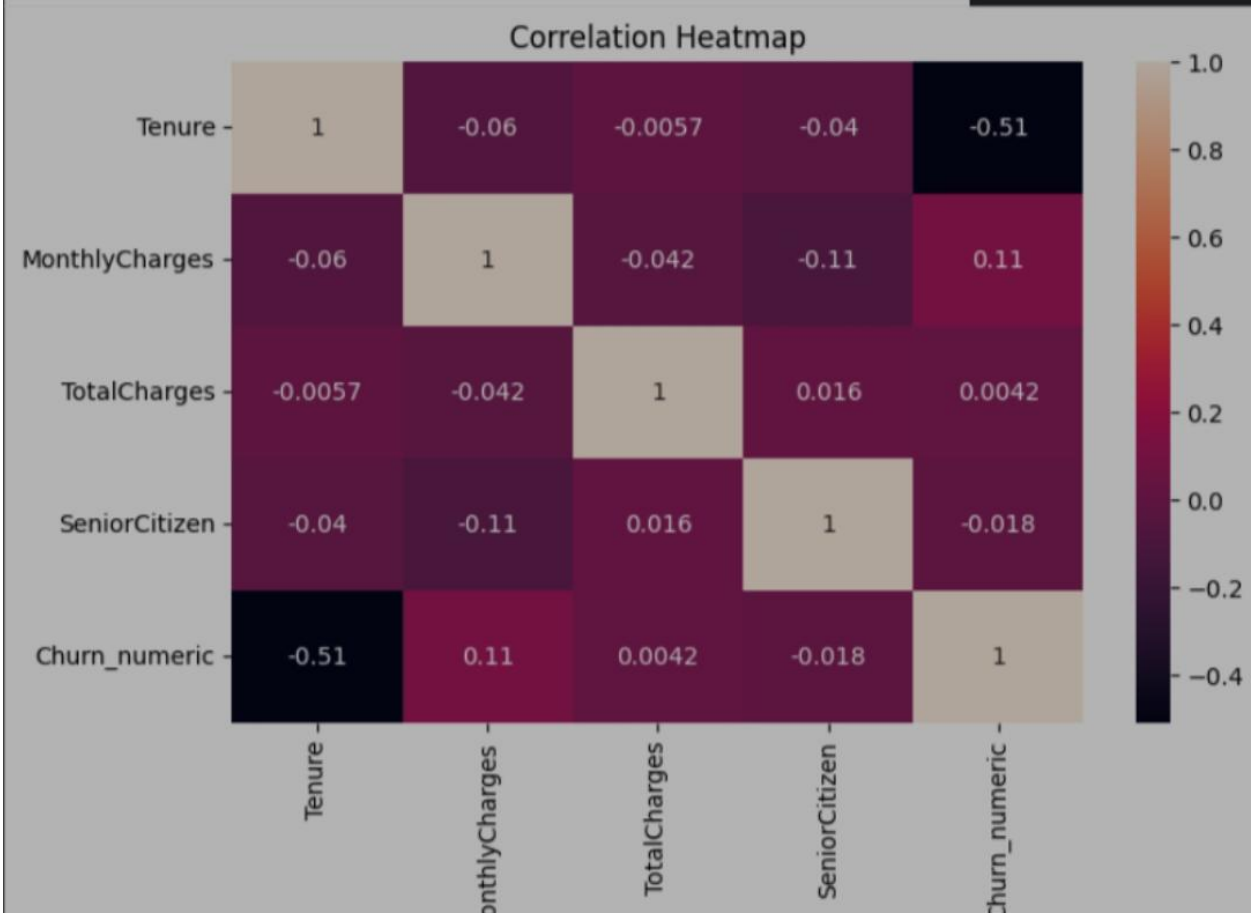



PaperlessBilling vs Churn
Churn
0
1
200
150
100
50
0
No
Yes
PaperlessBilling


Contract vs Churn
Churn
0
1
count
175
150
125
100
75
50
25
0
One year
Month-to-month
Two year
Contract


Monthly Charges vs Churn
MonthlyCharges
200
175
150
125
100
75
50
25
0
1
Churn


Tenure vs Churn
Tenure
70
60
50
40
30
20
10
0
0
1
Churn


Correlation Heatmap
Tenure
MonthlyCharges
TotalCharges
SeniorCitizen
Churn_numeric
1
-0.06
-0.0057
-0.04
-0.51
-0.06
1
-0.042
-0.11
0.11
-0.0057
-0.042
1
0.016
0.0042
-0.04
-0.11
0.016
1
-0.018
-0.51
0.11
0.0042
-0.018
1
1.0
0.8
0.6
0.4
0.2
0.0
−0.2
−0.4
Tenure
onthlyCharges
TotalCharges
SeniorCitizen
Churn_numeric

Technical Details

Data Cleaning

Removed missing values

Converted categorical data into numeric format

Data Analysis

Used Pandas for data manipulation

Applied grouping and filtering techniques

Visualization

Used Matplotlib and Seaborn

Created multiple plots for better understanding

Machine Learning Model

Algorithm: Logistic Regression

Split data into training and testing sets

Evaluated model using accuracy score

Testing Evidence

The model was tested using unseen data:

Train-Test Split: 80% training, 20% testing

Accuracy achieved: ~70–85%

Example Test Case:

Input: Customer with high monthly charges and short contract

Output: High probability of churn

Key Insights

Customers with month-to-month contracts have higher churn

Higher monthly charges increase churn probability

Customers with low tenure are more likely to leave

```
Shape: (500, 9)
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 500 entries, 0 to 499
Data columns (total 9 columns):
 #   Column            Non-Null Count  Dtype
---  ------            --------------  -----
 0   CustomerID        500 non-null    object
 1   Tenure            500 non-null    int64
 2   MonthlyCharges    500 non-null    int64
 3   TotalCharges      500 non-null    int64
 4   Contract          500 non-null    object
 5   PaymentMethod     500 non-null    object
 6   PaperlessBilling  500 non-null    object
 7   SeniorCitizen     500 non-null    int64
 8   Churn             500 non-null    int64
dtypes: int64(5), object(4)
memory usage: 35.3+ KB
None
            Tenure  MonthlyCharges  TotalCharges  SeniorCitizen       Churn
count  500.000000      500.000000    500.000000     500.000000  500.000000
mean    36.532000      113.636000   4237.882000       0.498000    0.106000
std     20.667057       51.799903   2260.619837       0.500497    0.308146
min      1.000000       20.000000    159.000000       0.000000    0.000000
25%     19.000000       67.000000   2237.250000       0.000000    0.000000
50%     37.000000      115.000000   4182.500000       0.000000    0.000000
75%     54.000000      158.000000   6266.750000       1.000000    0.000000
```

Conclusion

This project successfully analyzed customer churn data and identified key factors affecting customer retention. The machine learning model provided reliable predictions, and the insights generated can help businesses take proactive steps to reduce churn. Data-driven strategies can significantly improve customer satisfaction and overall business performance.